

Qingyang Liu

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Education

Duke University, School of Medicine, Durham, NC

Aug. 2024 – May 2026

Master of Biostatistics

University of Washington, Seattle, WA

Sept. 2020 – June 2024

BS in Applied Computational Mathematical Science

Experience

Biostatistics Research Analyst, Duke University Rohit Singh Lab

Oct. 2024 - Present

- Developed network-based gene prioritization framework integrating 33 biological networks across 13 categories
- Applied graph-base ranking algorithms with degree/density scoring to optimize disease-aware network selection
- Built differential expression re-ranking pipeline integrating scRNA-seq signals to improve person @10 by 100%
- Deployed Streamlit decision-support app enabling real-time rare disease gene prioritization for clinical use

Statistical Analyst, Duke Center For AIDS Research

Oct. 2024 - Present

- Constructed maximum-likelihood phylogenetic tree via RAxML-NG under GTR+Gamma model for 584 sequences
- Evaluated transmission cluster concordance using V-measure across 7 genomic subregions and 90+ parameter sets
- Benchmarked ClusterPicker and HIV-TRACE clustering methods across genetic distance and bootstrap thresholds
- Developed R pipeline for systematic multi-method, multi-threshold sensitivity analysis of partial-genome clustering

Algorithm Engineering Intern, Ping'an Technology

July 2023 – Sept. 2023

- Processed raw video data with FFmpeg and optimized parameters to improve quality for algorithm development
- Automated analysis of 80,000+ video files in Python, extracting metadata for database management and quality checks
- Built and organized databases containing 11,000+ minutes of processed data to facilitate AI training and evaluation

Projects

Pilot Trial Designing Software, Duke University

Aug. 2024 - Present

- Implemented a statistical framework to evaluate feasibility of pilot studies for future randomized clinical trials
- Applied Bayesian methods to estimate dropout probabilities, generate credible intervals, and assess study uncertainty
- Established composite decision rules combining acceptability and precision criteria to determine trial continuation
- Built an interactive Shiny app to calculate Bayesian assurance and support sample size planning for trial design

Multimodal Single-Cell Integration, University of Washington

Mar. 2023 – June 2023

- Built predictive models (Linear Regression, LGBM, MLP) to examine RNA–protein co-variation in single-cell dataset
- Trained models on 70,000+ CD34+ hematopoietic stem and progenitor cell samples collected from multiple donors
- Applied Python libraries (SCANPY, SHAP) to interpret outputs, quantify feature contributions, ensuring reproducibility
- Conducted single-cell RNA-seq analysis and generated visualizations to identify and communicate biological patterns

Vaccine Reservation System, University of Washington

Dec. 2021

- Developed SQL-based pipelines for hospitals and suppliers to update vaccine availability and distribution
- Simulated COVID-19 vaccine allocation strategies and supply chain management across pharmacies and hospitals
- Implemented user-level modules for appointment scheduling, access monitoring, and vaccine availability tracking
- Developed user-facing modules supporting secure login, appointment scheduling, and availability monitoring

Skills

Statistical Programming: R (tidyverse, Shiny), Python, SQL, Java

Statistical & Analytical Methods: A/B Testing, Bayesian inference, Regression Modeling, Exploratory data analysis

Healthcare & Bioinformatics Tools: Scanpy, SHAP, SEACells, scLink, Decipher, RaxML, HIV-Trace, TreeCluter

Research Tools: Git, Docker, LaTeX, FFmpeg